

NEP 2020 Optimized Timetable Constraints for Indian Higher Education System

HARD CONSTRAINTS (Mandatory)

- Timetable spans 5 working days per week.
- Lecture duration is 1 hour. Lab duration is 2 consecutive hours.
- Exactly one 1-hour break per day, not in the first or last slot.
- Division daily load must be between 6 to 8 academic hours.
- Weekly subject credit requirements must be satisfied as per curriculum.
- NEP credit structure must be satisfied: Major, Minor, Skill, Ability Enhancement, and Value Added credits.
- A subject can occur at most once per day per division.
- No teacher can be assigned to more than one session at the same time.
- Teacher daily workload must remain within institutional norms.
- No teacher may have more than 2 consecutive sessions.
- Open electives must not clash across divisions.
- Internship, field work, and teaching practice blocks must remain protected.
- Labs must occur every working day.
- Exactly 2 days must contain 2 labs; remaining 3 days must contain 1 lab.
- Each lab occupies 2 consecutive time slots.
- Divisions must be split into batches for labs, running simultaneously in different labs.
- Lab rooms cannot be double booked.
- Classrooms cannot be double booked.
- Lectures only in classrooms and labs only in labs.
- No division should have only theory sessions in a day; at least one lab or skill component required.

SOFT CONSTRAINTS (Optimization Goals)

- Avoid more than 2 heavy subjects consecutively for students.
- Maintain a daily mix of theory and lab or skill based sessions.
- Distribute teacher workload evenly across the week.
- Minimize idle gaps between teacher sessions.
- Avoid scheduling both earliest and latest slots on the same day for a division.
- Prefer labs before the final two slots of the day.
- Balance subject difficulty across the week.
- Maintain consistent elective slots across divisions.
- Maximize classroom and lab utilization efficiency.
- Ensure equitable daily academic load distribution across divisions.
- Preserve research blocks for faculty where possible.
- Encourage multidisciplinary exposure through subject distribution.